Within-fruit variability of reference soluble solids sets a ceiling on near-infrared calibration performance in apple

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ABSTRACT

Near-infrared (NIR) calibrations for fruit soluble solids content (SSC) and dry matter are judged by the root-mean-square error of prediction (RMSEP) against destructive reference determinations, yet the reference value of an intact fruit is a local sample of a heterogeneous organ. This study quantified how much of that error comes from within-fruit variability of the reference value and from between-instrument variability rather than from the model. Five faces of each of 633 apples were sampled destructively for SSC, and a public dataset of 4,318 kiwifruit scanned on 2–5 devices was analysed. Within-fruit variance was 52.30% of apple SSC variance; with face exchangeability not rejected and mean-independence assumed, this caps any whole-fruit predictor's R^2 at 0.477 (95% CI 0.436–0.515) for a single-face reference and 0.820 (0.794–0.842) for the five-face mean. Through the Spearman–Brown relation, the variance ratio gives a batch-calibrated lower bound on reference replicates per fruit and a parameter-free attenuation diagnostic, both presupposing leakage-free validation. Six parameter-free predictions were close to measurement; in a semi-synthetic demonstration with injected reference noise, the predicted decay held at all 12 finite replicate levels. Device and pooled fruit×device/scan-noise terms took 17.07% and 30.33% of scatter-corrected kiwifruit spectral variance (order-dependent shares); the pooled component showed almost no trend with model quality, so its share of mean squared error rose to 24.7%. Face-level random splitting inflated apple R^2 1.41–1.46-fold. Calibration performance should therefore be judged against a reference-limited ceiling, and the number of faces sampled per fruit planned in advance.

1. Introduction

Visible and near-infrared (NIR) spectroscopy has three decades of commercial use behind it for the non-destructive assessment of internal fruit quality attributes such as soluble solids content (SSC) and dry matter (DM) (Walsh et al., 2020). Portable and imaging instruments (Pissard et al., 2021; Wieme et al., 2022) and deep-learning models (Mishra et al., 2022; Garillos-Manliguez and Chiang, 2021) have widened its reach. Typical published figures for apple SSC lie at $R^2 > 0.85$ with a root-mean-square error of prediction (RMSEP) of order 0.5–1.0 °Brix (Walsh et al., 2020; Zhang et al., 2019; Li et al., 2022), with comparable levels for sugar content in other crops (Rady et al., 2021). However a calibration is built, its success is judged in the same way: a destructive chemical determination serves as the reference value, and RMSEP and R^2 on a prediction set are reported.

Much recent work has examined what limits the robustness of such calibrations. Mango DM calibrations have been validated on fruit of other cultivars and origin measured on another instrument (Mishra and Woltering, 2023), strawberry SSC calibrations on a cultivar absent from calibration and harvested at another time (Mishra et al., 2026), and for pear SSC, estimating fruit temperature alongside SSC improved prediction (Martins et al., 2023), while differences between instruments have motivated a large body of calibration-transfer methods (Mishra et al., 2021a; Mishra and Passos, 2021; Yang et al., 2022; Li et al., 2022). In strawberry, the RMSEP reduction obtained by fusing NIR and Raman spectra was largely associated with a decrease in bias (Mishra et al., 2026): the RMSEP of a calibration is a sum of parts, and the parts need not respond alike to a change in the model. In these studies the limiting factor is sought in the spectrum or in the sample population; the uncertainty of the reference value itself is not their object.

The reference value is itself a measurement. A destructive determination has a sampling location, a preparation protocol and an instrument reading, and therefore a variance of its own, and in intact fruit the sampling location matters because the constituents are not evenly distributed. The variation of SSC across different parts of pomelo fruit has been investigated in order to identify the optimal calibration position

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(Xu et al., 2024), and in thick-rind watermelon the uneven distribution of SSC may reduce the accuracy of quality analysis algorithms (Francis et al., 2025). Xiao et al. (2025) compared machine-learning regressions for citrus SSC built on single-position and on position-averaged spectra and found the averaged spectrum to give the better model, invoking uneven sugar distribution within the fruit as the reason position matters at all; Huang et al. (2026) likewise find the averaged spectrum to be their best-performing model for winter jujube, though not uniformly so across prediction sets, and treat measurement position as a source of prediction uncertainty to be suppressed. The theory of sampling states the general case: for heterogeneous materials the sampling error can exceed the analytical error of NIR analysis by factors of up to 25, depending on heterogeneity (Esbensen and Abu-Khalaf, 2022), and in non-destructive spectroscopic sensing of agri-food products sampling quality is the primary driver of overall uncertainty (Gajdoš Kljusurić et al., 2026). In these studies position is something to be chosen, averaged, predicted or suppressed. What is missing is the step from that observation to a budget: the within-fruit spread is itself an estimable variance component of the reference value, measured here on the reference rather than the spectral side, and under the variance model of Section 2.3 it sets a ceiling on the R^2 that a whole-fruit calibration can show against that reference.

Within chemometrics, the effect of reference error on calibration has a long history. Faber and Kowalski (1997) showed that validating against measured reference values makes the apparent prediction error systematically larger than the true one, because those values carry a component no model can predict, and proposed a correction for it. Sørensen (2002) applied such corrections to routine PLS calibrations: more precise reference data improved true accuracy only marginally when the reference error was already below the accuracy of the NIR calibration, and, where resources are limited, reference precision should be related to the accuracy the spectroscopic system can attain. A recent tutorial lists the effect of the uncertainty of the primary reference technique among the issues on which multivariate calibration still lacks a rational analysis, and on a public fruit dry-matter dataset it finds PLS prediction errors close to that uncertainty (Olivieri, 2026). Outside spectroscopy, errors-in-variables regression addresses error in the explanatory variable, where the consequence is a flattened slope (van Kippersluis et al., 2023); the disattenuation tradition instead removes the effect of unreliability from a correlation, and there the reliability of the response enters on the same footing as that of the predictor (Zhang and Wang, 2024). The error at issue here sits in the response alone and, under the classical additive model, leaves the slope unchanged but inflates the residual variance and lowers R^2. Certifying a reference material calls for the same decomposition and estimates it with the same difference of mean squares: replicate determinations within and between units, a one-way analysis of variance, and a homogeneity term that enters the uncertainty budget of the value finally assigned, which is the route Soulaimani et al. (2026) follow for a multi-element quinoa powder. Each of these analyses takes the size of the reference component as an input, and a replicate design is what supplies it. What that literature quantifies for a bottle of homogenised material can be quantified for an individual fruit, which cannot be homogenised before it is scanned, so a reference value assigned to it from m faces keeps a share of that variance. When the reference value is a single local determination, the reported R^2 cannot by itself tell a good model from a good reference method.

A second component that is not the model's doing is between-instrument variability. A large body of work has documented degraded cross-instrument performance and proposed correction methods (Mishra et al., 2021a; Mishra and Passos, 2021; Yang et al., 2022; Li et al., 2022), but that work concerns how to repair transfer, not how large the part of the prediction error that changes with the instrument is relative to the rest. On the spectral side, the covariance and correlation structure of measurement errors has been quantified across several units of the same miniaturised NIR instrument to guide preprocessing choices (Ezenarro et al., 2025), and measurement-error analysis has been applied to classification with handheld instruments (Ezenarro et al., 2024) and to the "error quality" of replicate scans (Riu et al., 2026). That work characterises the error structure of an existing measurement practice instead of proposing a new estimator; its counterpart on the reference side, where a value obtained by destructive sampling likewise has an estimable variance structure, is what this study supplies. A third issue is not a component of the budget but a precondition for computing it. An error budget partitions a reported RMSEP; if that RMSEP was inflated by a leaking validation split, the budget partitions the wrong total. Machine learning has documented how data leakage inflates reported performance (Kapoor and Narayanan, 2023; Rosenblatt et al., 2024), chemometrics has issued its own warnings (Király and Tóth, 2025), soil NIR calibration has quantified the overfitting produced when preprocessing such as principal component analysis or spatial interpolation is carried out before the split (Correa et al., 2025), and the optimistic bias of cross-validation is pronounced at limited sample sizes (Vabalas et al., 2019). Which grouping level a given dataset requires remains to be answered layer by layer and audited rather than assumed.

The aim of this study was to determine how accurately SSC and DM in intact fruit can be predicted from a spectrum when the destructive determination that

supplies the reference value carries its own uncertainty, by taking RMSEP apart into the parts that shrink as the model improves and the parts that stay essentially unchanged between the model operating points compared. Two structurally complementary datasets were used, one supplying replicate reference determinations (633 apples, five faces each) and the other replicate scans of the same fruit across instruments (kiwifruit scanned on two to five devices). The specific objectives were (1) to estimate the within-fruit variance of the reference SSC and the attainable R^2 ceiling it implies; (2) to confront the variance-component model with point predictions that contain no free parameters, and to invert it into the number of reference determinations per fruit needed for a target R^2; (3) to quantify the pooled between-instrument component of kiwifruit DM prediction error across model operating points; and (4) to audit the validation grouping level such estimates require, the individual level on the apple data and the acquisition-day level on a public time-series dataset. The design rule of Section 3.5 is meant to be used before an experiment: once the variance ratio σ_f^2/σ_a^2 has been estimated on a small subset of the target batch, inverting it gives a structural lower bound on the number of approximately independent reference determinations per sample needed for a target R^2 to be attainable under the variance model, the ceiling of Section 3.1 being a worked example rather than the end point. An attenuation-ratio diagnostic accompanies it, which any laboratory holding per-face (or per-point) replicate reference determinations can run on its own data.

The contribution is empirical and operational, not a new estimator or a new law. Variance components are long established in chemometrics; inverting a reliability into a required number of observations is the Spearman–Brown formula (Spearman, 1910; Brown, 1910), and asking how that reliability responds when observations on one facet are increased is what decision studies in generalizability theory do (Smith et al., 2025). The allocation principle is not new either: Sørensen (2002) related reference precision to attainable NIR accuracy, and the theory of sampling ties the number of increments a lot requires to its heterogeneity (Esbensen and Abu-Khalaf, 2022). What this study adds is the measurement the classical relation needs, made on destructive determinations of whole fruit, and the step that puts it to use as a planning rule whose output, the number of determinations per fruit for a target R^2, is a lower bound to be recalibrated on each batch. The variance-component model is required to deliver point predictions without free parameters, so that the decomposition can be refuted rather than merely described. No new regressor or preprocessing method is proposed. The intent is to give calibration design and performance assessment in non-destructive fruit quality testing a basis that can be computed in advance and rechecked on the reader's own data.

2. Materials and methods

2.1. Fruit samples and reference measurements

Two datasets carry the error budget (Table 1) in a relation of division of labour, not mutual corroboration: the per-face replicate structure unique to the apple data estimates the reference-side component, the cross-device replicate scans unique to the kiwifruit data the measurement-side component. Public datasets are typically "one fruit, one spectrum, one label" and structurally cannot yield the former.

The apples were bought in 2025 in a single acquisition from one wholesale market in Xinjiang, China, in three groups whose origins, Gansu, Shandong and Xinjiang, were declared by the merchants. Each fruit was sampled on five faces: the top, the bottom and three side faces on the equator, 120° apart. After spectral acquisition (Section 2.2), each face was sampled destructively and a digital refractometer (nominal accuracy ±0.2 °Brix) gave one SSC reading of its juice, so every fruit carries five separate reference determinations. A cleaning rule fixed before analysis removed whole fruit containing any face above 20 °Brix, the physiological ceiling being 18–20, and kept values below 5 °Brix, since unripe or rotten tissue genuinely reaches 3–5 and removing them would shrink the finding a priori. After cleaning, 633 fruit × 5 faces = 3,165 reference determinations remain, a fully balanced design used for the variance-component analysis; 3,149 of them have a matching spectrum and are used for spectral modelling.

The kiwifruit data are public, on figshare under CC BY 4.0 (Wohlers, 2025), first described by Wohlers et al. (2023), and were used exactly as deposited. They give SSC and DM reference values for 5,418 fruit and 11,982 spectra; 4,318 fruit were scanned by two to five devices (only 90 by all five), and the further 1,100 fruit scanned once contribute nothing to the instrument component.

A third dataset appears once and is declared here so that the count is not misread: the public DeepHS-Fruit time-series hyperspectral set of post-harvest ripening sequences introduced by Varga et al. (2021), in its 2023 five-species release (3,405 VIS spectra), used in Section 3.4 and the Supplementary Material. It contributes nothing to either variance component and to no claim the paper makes; it is the dataset on which we initially read a day-level leakage effect and then withdrew that reading, kept because the trap it illustrates is worth reporting.

2.2. Spectral acquisition

Apple spectra were acquired in reflectance mode with a visible–NIR hyperspectral imaging system over 391–1001 nm (346 bands), several fruit per image, and

Table 1
Data card. The two datasets carry different component-estimation tasks in this paper.

	Apple (reference-side component)	Kiwifruit (measurement-side component)
Source	Collected by Xinjiang Agricultural University in 2025; previously unpublished; not publicly released under the holding institution's data policy, available from the corresponding author on request	Public, on figshare under CC BY 4.0 (Wohlers, 2025), first described by Wohlers et al. (Wohlers et al., 2023); used exactly as deposited, with no augmentation from that work
Origin / batch	Merchant-declared origins Gansu, Shandong and Xinjiang; all three groups purchased in one 2025 acquisition from a single wholesale market in Xinjiang	Origin not distinguished
Sample size	633 fruit $\times$ 5 faces = 3,165 reference determinations after cleaning; 3,149 rows remain after fruit–face pairing (16 faces have no matching spectrum). The variance-component analysis uses the former (fully balanced), spectral modelling the latter	5,418 fruit / 11,982 spectra
Key replicate structure	Five faces per fruit: top, bottom and three equatorial side faces $120°$ apart	Cross-device replicate scans: 4,318 fruit scanned by 2–5 devices (only 90 by all five); the further 1,100 scanned once contribute nothing to the instrument component
Reference determination	After destructive sampling, a digital refractometer gives one reading per face (nominal accuracy ± 0.2 °Brix)	SSC and dry matter (DM)
Spectra	Reflectance imaging, 346 bands over 391–1001 nm; archived after min–max scaling of each spectrum to $[0, 1]$	222 bands over 402–1065 nm, the common usable range of the five devices
Cleaning rule (fixed before analysis)	Remove whole fruit containing any face with SSC $>$ 20 °Brix (the physiological ceiling is 18–20, so higher values are entry errors); keep values < 5 °Brix, since unripe or rotten tissue genuinely reaches 3–5 and removing them would shrink the finding a priori	Drop rows with missing values and the wholly missing 1068–1137 nm columns, absent on all five devices, leaving 222 bands; a later missing-channel mask removes 0 of those 222. The instrument-side analysis uses only the subset scanned by ≥ 2 devices
Use in this paper	Within-fruit reference variability, attainable ceiling, $1/m$ design rule, attenuation diagnostic	Between-instrument error budget, comparison of two model operating points
SHA-256 (first 16)	`b5e923b4d05be65f` (cleaned analysis table; the two raw workbooks are `e77fb341681ac5e7` and `bfdbeba89ef79b80`)	`37157f2c42a5d441` (original deposit as downloaded)

one spectrum was archived for each fruit face. The archived spectra had been min–max scaled to $[0, 1]$ spectrum by spectrum, so their absolute reflectance cannot be recovered and detector saturation cannot be read from their values; a plateau of ten or more bands tied at the maximum, the signature of clipping, occurs in 33 of the 3,228 archived spectra (1.0%). "Raw" below refers to these scaled spectra. The kiwifruit spectra come from five devices; the 222 bands over 402–1065 nm that form their common usable range are used, after the cleaning given in Table 1.

2.3. Variance-component model

For the per-face reference values a one-way random-effects model is used

$$\begin{aligned} &Y_{ij} = \mu + a_i + e_{ij}, \\ &a_i \sim (0, \sigma_a^2), \quad e_{ij} \sim (0, \sigma_f^2), \quad j = 1, \ldots, 5, \end{aligned} \tag{1}$$

where a_i is the whole-fruit deviation of fruit i and e_{ij} the deviation of its j-th face from that fruit's own level $\mu + a_i$. The design is balanced, so the classical ANOVA moment estimators $\hat{\sigma}_f^2 = \mathrm{MS}_{\mathrm{within}}$ and $\hat{\sigma}_a^2 = (\mathrm{MS}_{\mathrm{between}} - \mathrm{MS}_{\mathrm{within}})/5$ are used rather than REML, which keeps the decomposition directly recheckable from the mean squares (Liljequist et al., 2019) but still relies on the additivity and homoscedasticity assumptions of the model. The ICC, the minimal detectable difference derived from it and agreement analysis in method comparison are established measurement tools, as are the known limitations of judging agreement by a correlation coefficient (Schiano-Lomoriello et al., 2021; Gerke, 2020; Janse et al., 2021). Intervals for the variance components and the ICC come from bootstrap resampling of fruit, the five faces moving together ($B = 2000$).

If the reference value is the mean of m faces, $\bar{Y}_i^{(m)}$, then any model that can perceive only whole-fruit information (i.e. can predict only $\mu + a_i$) satisfies

$$R^2 \leq \frac{\sigma_a^2}{\sigma_a^2 + \sigma_f^2/m}, \qquad \mathrm{RMSE} \geq \frac{\sigma_f}{\sqrt{m}}. \tag{2}$$

Equation (2) constrains the population R^2, whereas the R^2 reported here is a finite-sample score referenced to the training-fold mean and so never lower than under the test-mean convention (Section 2.4), which leaves it not guaranteed to obey the bound exactly in finite samples. Every comparison drawn here between a measured score and this ceiling, and equally the six point predictions of Section 3.2 and the design rule of Section 3.5, is approximate to the order quantified in Supplementary S2 (a mean of 0.004–0.006 in R^2 on the apple data, where every R^2 is small, and 0.0002 on the kiwifruit data), and the bound is not claimed as an exact property of the reported statistic.

Two conditions are required, and it is worth separating them. The first is exchangeability of the five faces:

the deviations e_{ij} are mutually uncorrelated within a fruit, so the variance of their mean is exactly σ_f^2/m. The second is that the face deviations carry no information the predictor can see. Zero covariance, $\mathrm{Cov}(a_i, \bar{e}_{i\cdot}) = 0$, does not suffice, because it constrains only linear dependence whereas the cross-term that must vanish when a predictor g is subtracted is $\mathbb{E}[(a_i - g)\,\bar{e}_{i\cdot}]$, which for nonlinear g a covariance does not control. The condition stated here is the stronger one: the mean deviation $\bar{e}_i^{(m)}$ over the m faces serving as the reference value is mean-independent of every whole-fruit quantity a predictor can condition on, $\mathbb{E}[\bar{e}_i^{(m)} \mid a_i, S_i] = 0$, with S_i the spectrum, and it is required separately for each m in use. Three explicit constructions in Supplementary S1 show why each strengthening is needed and that under zero covariance alone Eq. (2) holds only against predictors affine in the whole-fruit information. Both conditions are assumptions rather than consequences of writing the model down. The first is tested in Section 3.2; the second is not testable on this dataset in either version, because it would need technical replicates on the same face (Section 4.3), and it is imposed. Under merely approximate mean independence the expression is an approximation whose error is of the order of the residual predictable part of $\bar{e}_i^{(m)}$.

The m "replicates" here are destructive samplings at m different positions on the same fruit, not m replicate analyses of the same sample, so σ_f^2 is the spatial-subsample variability defined by the five-point scheme, pooling between-face chemical heterogeneity, sampling and preparation variability, and refractometer reading error, which are inseparable here. Equation (4) accordingly asks how many approximately independent spatial subsamples per fruit are needed, not how many parallel determinations the reference method must perform; Supplementary S1 sets out the distinction and records that the five faces were the pre-existing scheme of this collection, so the $1/m$ test of Section 3.2 is an in-sample check.

2.4. Calibration modelling and validation

Unless stated otherwise, partial least squares regression (PLS) is used, with the component count selected by grouped cross-validation within the training set; every split behind a reported figure is fruit-disjoint, and the row-wise splits of Section 3.4 are leakage controls rather than estimates. Using one regressor is deliberate. Eq. (2) bounds every predictor that perceives only whole-fruit information and Eq. (4) inherits that property, so neither belongs to the regressor: a stronger learner can carry a study closer to the ceiling but cannot raise it, and a model comparison would vary the distance to the bound while leaving the bound itself, the object measured here, untouched. PLS fills the role because it is the reference method of spectroscopic calibration (Mishra et al., 2021b, 2022) and its latent structure reduces tuning to a single decision, the component count. Where the restriction costs something it is stated: other model classes and parameter spaces were not individually excluded, so the cause of the low apple-side performance remains a candidate rather than a demonstration (Section 4.1). The "weak model" operating point of Section 3.3 is a deliberate exception, its component count fixed at 10 to construct a poorly performing comparison point. Preprocessing compares raw, SNV, Savitzky–Golay first derivative and their combinations (Mishra et al., 2021b, 2022), following published best-practice guidance (Burnett et al., 2021) and comparable quantitative applications (Meacham-Hensold et al., 2019): scatter correction and derivative filtering address the two artefacts that dominate diffuse reflectance on intact fruit, multiplicative path-length variation and baseline drift, and are compared on the data rather than assumed.

The total sum of squares in R^2 is referenced to the training-set mean rather than the test set's own mean, because the training mean is the only baseline available at prediction time. An algebraic identity makes the training-mean denominator never smaller and the resulting R^2 never lower; replaying every split used here without refitting, the two conventions differ by 0.0002 in R^2 on the kiwifruit data and by a mean of 0.004–0.006 on the apple data, where every R^2 is below 0.12, and no point prediction or interval conclusion changes under the other convention. The identity, the measured denominator ratios and the converted attenuation test are in Supplementary S2.

Except for the row-wise random split used deliberately as a leakage control in Section 3.4, and for the acquisition-day grouping described next, every performance estimate is grouped by individual: multiple spectra of the same fruit never straddle the train/test boundary. For the time-series spectra, grouping by acquisition day is evaluated as a second grouping level parallel to the individual one, not a further layer on top of it: it holds out whole acquisition days and does not block the individual level, so spectra of the same fruit taken on different days do fall on both sides of the boundary. The two levels are audited separately, and neither estimate is a generalization estimate for unseen fruits and unseen acquisition days at once.

Replication and uncertainty reporting. Everything depending on a random split runs in two stages: feasibility is verified on a single development seed (Pilot), then the whole analysis is rerun on five seeds fixed in advance (Formal), and every split-dependent quantity in the text is the Formal result. A second class depends on no random split, namely the variance components and ICC, the attainable ceiling, the label-only baseline and the spectral-quality and band-extreme diagnostics; these are seed-independent, so the original computation was retained. Intervals for split-dependent quantities use a two-level cluster bootstrap with seeds as clusters,

because repeats under one seed share a single random stream and an i.i.d. bootstrap underestimates interval width; the reported SDs reflect only splitting and modelling variance, not hardware jitter. Every ratio-valued quantity is computed as a ratio of means rather than a mean of ratios, because predictions such as Eq. (3) are themselves ratios of expectations. The seed list, the repeat count of every analysis, the bootstrap construction and the measured gap between the two ratio conventions are in Supplementary S2.

Computation runs on CPU. Given the seed, a rerun on the same architecture and BLAS implementation returns the reported values bit for bit; across architectures it does not, because the inner cross-validation MSE curve is nearly flat in weak-signal configurations, so floating-point differences can move the component selection. A full replay on a second architecture, reported in Supplementary S2, leaves most checked quantities unchanged and reverses no statistical conclusion. The dependence is removable rather than merely disclosed, since the component count selected for every configuration is released with the model records and holding it fixed reproduced one repetition of a checked run to the sixth decimal place on the other architecture (a single check, not one per configuration). The software environment that produced the reported numbers was not recorded at the time; the pinned environment released with the code is the one in which the replay was run, and is labelled as such.

2.5. Pre-registration and traceability

The statistical analysis plan, the cleaning rules and every criterion were archived before the final results were seen. Every variance component, R^2, RMSE, ratio and confidence interval reported here is produced by a script and written to a workbook, and each such number in the figures is traceable to a specific table; dataset metadata are traceable to the data card instead. Quantities the text reports but derives from other stored components are themselves exported, so that "traceable" means stored, not merely recomputable. Process-level intermediates such as the bootstrap resamples can be replayed from the scripts and seeds. Any deviation from the analysis plan is entered in the protocol's deviation log.

3. Results

3.1. Within-fruit variability of reference SSC and the attainable ceiling

After cleaning, $n = 633$ fruit, $\hat{\sigma}_a^2 = 2.327$ and $\hat{\sigma}_f^2 = 2.551$: the within-fruit (between-face) variance is 52.30% of the total (ICC = 0.477, bootstrap 95% CI [0.436, 0.515]). The within-fruit SD is $\hat{\sigma}_f = 1.597$ °Brix and the standard error of the five-face mean $\hat{\sigma}_f/\sqrt{5} = 0.714$ °Brix; the SD of the single-face values is larger, 2.209 °Brix, because it also carries $\hat{\sigma}_a$, and the two must not be confused. The median within-fruit range is 2.80 °Brix, 44.1% of fruit exceed 3 °Brix and 20.7% exceed 5 °Brix (Fig. 1a); the median within-fruit SD across five faces, 1.110 °Brix, is 5.6 times the refractometer's nominal accuracy.

The proportion differs markedly between the three origin groups: 38.5% for Shandong, 52.5% for Xinjiang and 66.5% for Gansu (Fig. 1b), all bought in the same acquisition from the same wholesale market (Section 2.1). No hypothesis test of the between-group differences is performed; these are point estimates only. The corresponding attainable R^2 ceilings run from 0.615 down to 0.335, nearly a factor of two, so the ceiling is not a species constant and must be reported per collection.

By Eq. (2), a single face as the reference value gives an R^2 ceiling of 0.477 and the mean of five faces 0.820 (Fig. 1c), both point estimates quoted as such throughout. Propagating the ICC interval gives [0.436, 0.515] for a single face and [0.794, 0.842] for the five-face mean; the conservative reading, under which a model is hardest to declare ceiling-bound, takes the upper limits 0.515 and 0.842, and every statement about this ceiling should be read as carrying that interval.

Part of the within-fruit variability could simply be the known calyx–stem gradient. The gradient is confirmed, the bottom exceeding the top by 0.405 °Brix (paired $t = 4.42$), but it is only 36% of the median within-fruit SD and the fixed effect of face position explains just 0.95% of the within-fruit variability. Restricting to the three side faces, which lie on the same equatorial band and carry no stem–calyx gradient, still leaves a between-face variance of 2.4396, i.e. 95.6% of the value 2.5515 estimated from all five faces. The variability therefore has almost no structure in the spatial variables this dataset records, which is not to say it has none, since it may relate to unrecorded factors such as sun exposure or canopy position; for that reason standardising the sampling position alone will not eliminate it. Spatial dynamics of sugars and cell state within apple flesh have direct cytological evidence in watercored fruit (Wada et al., 2021), a physiological disorder, so that evidence is compatible with the dispersion seen here but is not extended to sound fruit, and with no technical replicates on the same face (Section 4.3) this dataset cannot attribute the dispersion to biological heterogeneity either. The full control is in Supplementary S5.

3.2. Parameter-free predictions of the variance-component model

Equation (2) and its corollaries give four groups totalling six parameter-free point predictions checkable against the data (Fig. 2; all values are in the companion workbook).

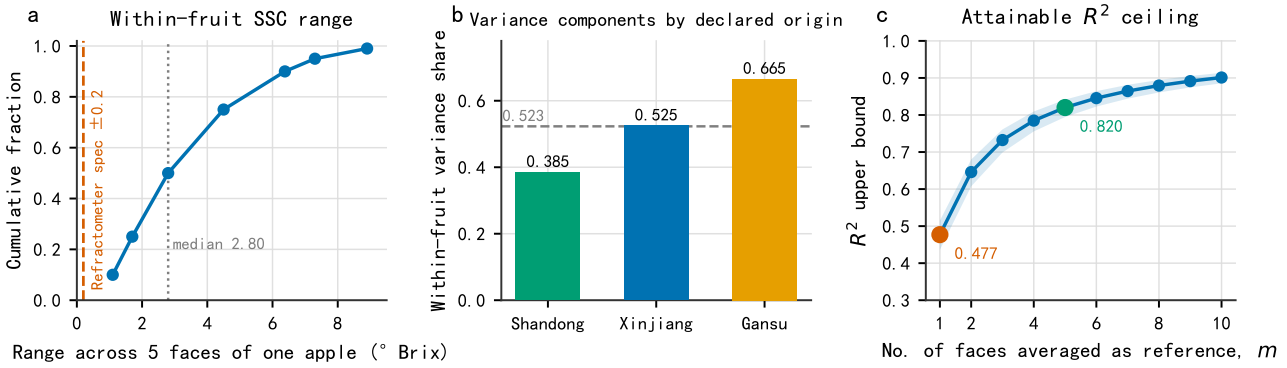


Figure 1: Within-fruit variability of the reference value. (a) Cumulative distribution of the SSC range across the five faces of a fruit; the dashed line marks the refractometer's nominal accuracy and the dotted line the median range. (b) Within-fruit share of the total variance by origin group; the horizontal dashed line is the pooled value 0.523. (c) Attainable R^2 ceiling against the number of faces m averaged into the reference value; the shaded band is the bootstrap 95% interval of the ICC.

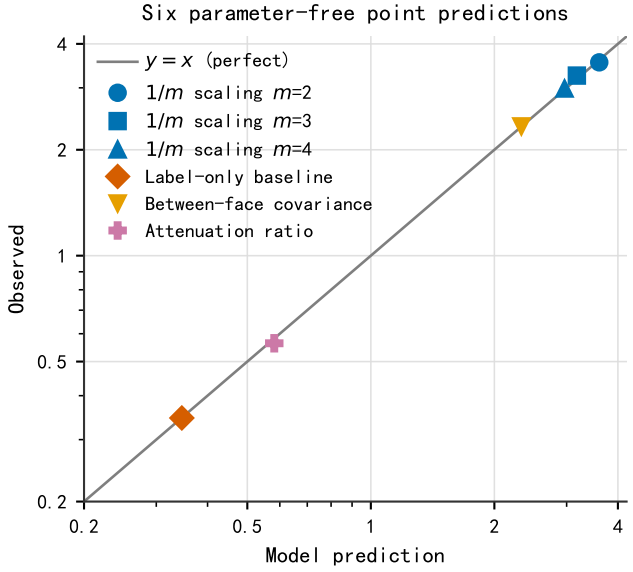


Figure 2: Parameter-free point predictions: abscissa, the value predicted by the variance-component model; ordinate, the measured value; grey line, $y = x$.

The first group is the $1/m$ scaling. Taking the mean of the first m sampling faces of each fruit in a fixed order, the measured variance deviates from the theoretical value $\sigma_a^2 + \sigma_f^2/m$ by -1.61% ($m = 2$), $+2.30\%$ ($m = 3$) and $+0.72\%$ ($m = 4$). Both variance components are estimated once from the complete $m = 5$ design, so the test is not circular; a fixed subset is used rather than the average over all $\binom{5}{m}$ combinations, so sensitivity to subset choice is untested, and this remains an in-sample extrapolation.

The second group tests exchangeability across faces. The mean off-diagonal element of the covariance matrix is 2.331, consistent with the value $\hat{\sigma}_a^2 = 2.327$ predicted under exchangeability, and a permutation test of compound symmetry cannot be rejected ($p = 0.261$). Failure to reject is not proof that the scaling holds exactly.

The third group is the label-only baseline. Predicting the SSC of one face from the mean of the other four faces of the same fruit has theoretical value $1-(\sigma_f^2+\sigma_f^2/4)/(\sigma_a^2+\sigma_f^2) = 0.3462$ against a measured 0.3456. This expression contains no free parameter, is the sharpest of the group, and involves no train/test split, so the denominator-convention note of Section 2.3 does not apply to it.

The fourth group is the attenuation ratio, the only one carrying a confidence interval, treated separately below.

Suppose a model perceives only whole-fruit information and extracts the same signal under both targets, so that the variance it explains is the same for both; the

ratio of R^2 between the targets then equals the ratio of their variances. The second half of that supposition is an assumption in its own right, since two predictors can both be confined to whole-fruit information and still fit different amounts of it; the paired design below makes it plausible and the interval test would expose it if it failed. One subtlety decides the ratio: the "whole-fruit" target is the mean of five faces, so it still carries noise $\sigma_f^2/5$, and the prediction is not the ICC but

$$\frac{R^2_{\text{face-level}}}{R^2_{\text{whole-fruit mean}}} = \frac{\sigma_a^2 + \sigma_f^2/5}{\sigma_a^2 + \sigma_f^2} = 0.5816. \qquad (3)$$

It is tested under a paired design in which the two targets share the same splits, the numerator and denominator being the raw-preprocessing face-level and fruit-mean R^2 values of the main apple table; the corresponding figures in the leakage controls of Section 3.4 come from a separate paired run with its own splits and can neither be interchanged with these nor used to recompute the ratio. The measured ratio is 0.564, with a two-level seed-clustered bootstrap 95% confidence interval [0.531, 0.596] that contains the prediction 0.5816. The comparison is discriminating rather than merely permissive: the ICC itself (0.477, which corresponds to a noise-free whole-fruit target, i.e. the limit $m \to \infty$) falls outside that interval, so the data distinguish between two candidate predictions differing by only 0.10 and select the one the variance-component model implies. Both numerator and denominator are R^2 values of magnitude only 0.06–0.12, so the ±0.03 width is driven mainly by the weakness of the spectral signal rather than by unstable splitting, the margin by which the interval clears the ICC is correspondingly modest, and nonparametric bootstrap intervals have been reported to carry inflated type I error rates under some conditions even where their power is good (Madssen et al., 2025), so a margin this narrow is provisional rather than decisive. This discrimination is therefore the weakest of the point predictions and should be rechecked where the spectral signal is stronger.

3.3. Between-instrument variability in kiwifruit dry matter prediction

In the kiwifruit data there are no technical replicates within a device, so the instrument×sample interaction and the scan/repositioning noise are inseparable and only their sum can be reported. On that basis the variance of the SNV-corrected spectra decomposes into 52.60% between fruit, 17.07% device main effect and 30.33% fruit×device pooled residual (Fig. 3a): nearly half of the spectral variation originates in the measurement system rather than in biological difference. The device main effect is a systematic bias, identifiable under a crossed design and in principle correctable, which is precisely the target of calibration-transfer methods (Mishra et al., 2021a; Yang et al., 2022; Li et al., 2022).

The predictive-level result is more direct. At two model operating points, a weak model (SNV, full spectrum, 10 components) gives $R^2 = 0.393$ and RMSEP = 1.809 %DM and a strong model (first derivative, 700–1065 nm, cross-validated component count) $R^2 = 0.819$ and RMSEP = 0.988 %DM. Their RMSEPs differ by a factor of 1.83, but the part of RMSEP corresponding to the dispersion of predictions for the same fruit across devices barely moves: 0.495 versus 0.490 %DM. Two points cannot distinguish invariance from a shallow trend, so the operating range was scanned. The scan uses a different fold rule, assigning fruit at random to five folds under each fixed seed rather than the deterministic grouped k-fold used above, because a slope needs an uncertainty; its numbers are therefore its own and are not interchangeable with the two above. Ten configurations crossing five preprocessing methods, three spectral ranges and five component budgets were fixed in advance, the two operating points above being two of them; all ten are listed in Table S1 of Supplementary S3. They span R^2 from 0.097 to 0.820, over which the total RMSEP falls by 55% (2.207 to 0.986 %DM).

Across that scan the pooled component stays between 0.457 and 0.562 %DM (mean 0.505), while its share of mean squared prediction error rises from 4.3% to 24.7%, a factor of 5.7 (Fig. 3b). Regressing the component on R^2 gives a slope of +0.0073 %DM per unit R^2, with a seed-clustered bootstrap 95% interval of [+0.0038, +0.0109] that excludes zero, so the component is not strictly invariant and that is reported rather than invariance claimed. The size of the trend is what matters. The regression accounts for 0.5% of the variance of the component across configurations, and over the entire R^2 range the fitted trend predicts a change of +0.005 %DM, i.e. 1.1% of the component's own magnitude, while the denominator it is divided by shrinks by more than half. That 1.1% is the trend, not the observed movement: the measured values span 20.6% of their mean across the ten configurations, and the residual SD of ±0.03 %DM shows that almost all of that spread is configuration-to-configuration scatter unrelated to model quality. All five fitted diagnostics are exported alongside the slope.

Two qualifications stand: the quantity is the pooled device×sample interaction and scan/repositioning noise rather than pure instrument repeatability, and the configurations differ simultaneously in preprocessing, spectral range and component budget, so the scan is a sweep of operating points rather than a controlled ablation. Within those qualifications, model improvement compresses the compressible part of the error and leaves the pooled component very nearly untouched, so the share of the latter rises steeply.

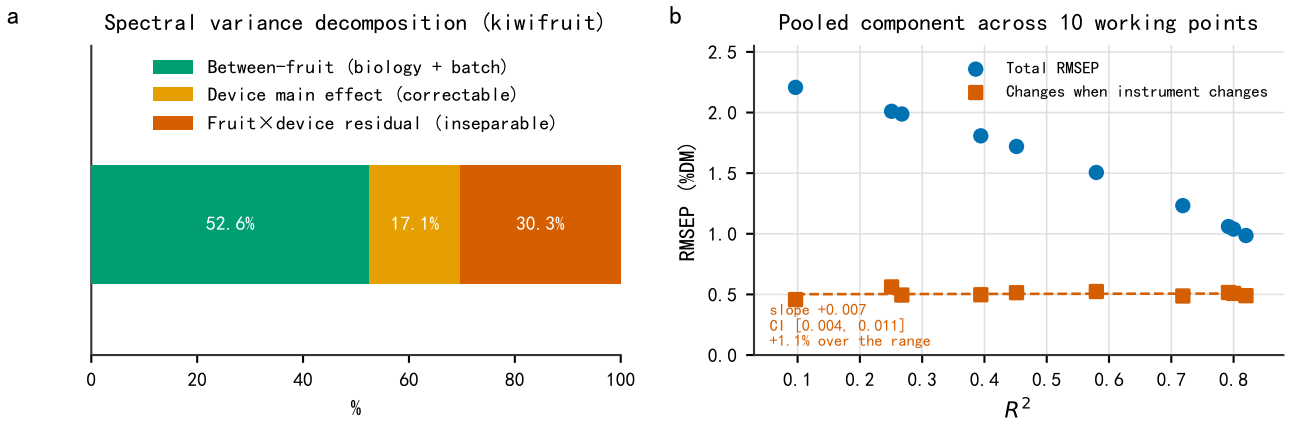


Figure 3: Between-instrument variability on the measurement side. (a) Three-component decomposition of the SNV-corrected spectral variance. (b) Total RMSEP (blue circles) and the pooled component that varies across devices (orange squares, being device×sample interaction together with scan/repositioning noise, which this design cannot separate) at ten pre-specified operating points, against R^2 on the abscissa. The dashed orange line is the linear regression of that component on R^2 and the band is its seed-clustered bootstrap 95% interval.

3.4. Validation grouping level and data leakage

The components above are meaningful only if the validation protocol itself is leakage-free. Cross-validation practice and nested feature selection have been discussed systematically in the methodological literature (Wilimitis and Walsh, 2023; Lewis et al., 2023); what this section adds is that the grouping level itself must be determined layer by layer from how the data were generated. It reports the one leakage confirmed here; a second, day-level reading withdrawn after a control test is documented in full in the Supplementary S4, because the trap it illustrates is easy to miss.

In the apple data, replacing a row-wise random split by grouping by fruit lowers R^2 from 0.0878 to 0.0625 (face-level SSC) and from 0.1692 to 0.1195 (whole-fruit mean SSC), inflation factors of 1.41 and 1.42; under first-derivative preprocessing the same comparison gives 0.0917 → 0.0631 and 0.1580 → 0.1079, factors of 1.45 and 1.46, so across the four combinations the inflation is 1.41–1.46×. The seed-clustered bootstrap 95% confidence intervals are [1.21, 1.71], [1.27, 1.63], [1.25, 1.67] and [1.30, 1.71], all four lower limits sitting clearly above 1 (Fig. S2 of the Supplementary Material). The mechanism is that the remaining faces of the same fruit fall into the training set and leak whole-fruit information.

On the time-series hyperspectral data this work initially read a second, deeper layer of leakage: predicting the imaging day index of a single spectrum gives $R^2 = 0.887$ when grouped by fruit and 0.705 when grouped by acquisition day. That reading does not survive a control test and we withdraw it. The day index is numbered from the acquisition-day label, so holding out a whole day also removes an entire level of the target; rerunning with destructive measurements not derived from the grouping variable leaves only 2 of 6 combinations above the pre-specified threshold, both on the one target still strongly correlated with acquisition day. We therefore do not claim a day-level mechanism independent of individual-level leakage. The episode is nonetheless a reproducible protocol trap: when the target is derived from the grouping variable, grouping by it manufactures a leakage-like drop out of nothing, and the size of the drop alone cannot distinguish the two. Supplementary S4 gives the full analysis, the terminal-date decomposition and Fig. S1.

3.5. Number of reference determinations required per fruit

Inverting Eq. (2) answers how many independent replicate determinations the reference value must have for the attainable R^2 ceiling to reach a target $R^2_{\max}$:

$$m \geq \frac{\sigma_f^2}{\sigma_a^2} \cdot \frac{R^2_{\max}}{1 - R^2_{\max}}. \tag{4}$$

Equation (4) is the Spearman–Brown prophecy formula solved for the number of replicates (Spearman, 1910; Brown, 1910), as Supplementary S1 shows term for term. What this section adds is empirical and structural rather than algebraic: a σ_f^2/σ_a^2 measured on near-infrared calibration material, the target reliability read as an attainable R^2 ceiling, and the inversion written as a point prediction carrying no free parameters. Section 3.6 examines the related degradation prediction, and a blinded inversion of it, under injected reference noise.

Substituting the apple value $\sigma_f^2/\sigma_a^2 = 1.0965$ gives the design table: $R^2_{\max} = 0.7$ requires $m \geq 3$; 0.8 requires $m \geq 5$; 0.9 requires $m \geq 10$; 0.95 requires $m \geq 21$. Point estimates of the ratio already differ

between the three origin groups of this collection, although the differences were not tested (Section 3.1), so practical use should first estimate σ_f^2/σ_a^2 on a small subset of the target batch; batch membership is not recorded for this material, so no claim is made about how the ratio changes with batch. Equation (4) gives a structural lower bound, not a sufficient condition: in the semi-synthetic test of Section 3.6 the measured R^2 falls slightly short of the parameter-free prediction, partly through the further loss incurred when a model is retrained on noisy reference values. A margin above the bound is therefore advisable: the advisory rule, an empirical margin induced from four blinded targets and to be recalibrated on other data (Section 3.6), is to discount the clean baseline R^2_{clean} by 1% before inverting, a discount factor that is not the same quantity as the relative deviations reported there.

3.6. Semi-synthetic demonstration of reference-noise degradation

Sections 3.1–3.5 all rest on the apple data, whose spectral side performs poorly under the PLS protocols evaluated here, and a ceiling never approached by any working model might be a property that is never triggered. A semi-synthetic experiment was therefore run, whose criteria were archived before it was run and whose four design targets were fixed before the corresponding replicate levels had been measured. This is the second version of that pre-registration; the first is kept verbatim rather than deleted, and Supplementary S5 records what it set wrongly and that no threshold was changed in correcting it.

The analysis moves to the kiwifruit data, whose spectral signal is strong enough (calibration R^2 of 0.827 for DM and 0.861 for SSC), and injects reference noise into the labels: for each fruit the original label is replaced by the mean of m virtual replicate determinations, adding independent noise of variance σ_f^2/m. The kiwifruit label variance σ_y^2 is split into a between- and a within-fruit part in the ratio $\sigma_f^2/\sigma_a^2 = 1.0965$ measured on apples, and the within-fruit part is injected, $\sigma_f^2 = \left[1.0965/(1+1.0965)\right]\sigma_y^2 = 0.5230\,\sigma_y^2$, exactly the 52.30% within-fruit share of the apple data ($= 1 - \text{ICC}$); the split is stated rather than the ratio alone because the alternative reading $\sigma_f^2 = 1.0965\,\sigma_y^2$ differs by a factor of 2.1 and does not reproduce the required replicate counts of the blinded test below. The model is retrained on the noisy labels, the component count also being selected using them, which corresponds to the real situation in which the experimenter only ever holds noisy reference values. Thirteen levels of m are used $(1, 2, \ldots, 10, 14, 20, \infty)$ with the Formal five seeds, under the fruit-grouped repeated holdout that produces the values 0.827 and 0.861; that is not the 5-fold grouped cross-validation giving $R^2 = 0.819$ in Section 3.3, and the two must not be mixed.

Under this setup the model can perceive the entire clean label (variance σ_y^2) and only the injected σ_f^2/m is imperceptible to it, so the structural ceiling of Eq. (2) becomes $\sigma_y^2/(\sigma_y^2+\sigma_f^2/m)$ and the parameter-free prediction is

$$R^2_{\text{pred}}(m) = \frac{R^2_{\text{clean}}\,\sigma_y^2}{\sigma_y^2 + \sigma_f^2/m}. \tag{5}$$

Carrying R^2_{clean} through unchanged is exact for a predictor fixed before the injection, whose error is then uncorrelated with the injected noise; the learner here is refitted on the noisy labels and selects its component count with them. Fixing the components and coefficients on the clean labels instead raises R^2 at all twelve finite levels, by 0.0039 at $m = 1$ falling to 0.0002 at $m = 20$, so Eq. (5) is optimistic for the retrained procedure by that much, which is the conservative direction for a design rule (Supplementary S5).

At none of the 13 levels of m does the measurement cross the structural ceiling (Fig. 4). That is an arithmetic consequence of the clean baselines rather than an identity: the pre-registered tolerance $|R^2_{\text{obs}}/R^2_{\text{pred}} - 1| \le 15\%$ is narrower than the factor R^2_{clean} by which R^2_{pred} already sits below the ceiling. The observation is consistent with the declared injected variance but does not by itself establish it, since staying below the ceiling is compatible with more than one injected magnitude; what the injection realises rests on the implementation and the pre-registered parameters, as Supplementary S5 sets out. The median relative deviation between measurement and Eq. (5), over the twelve finite-noise levels, is −0.9% (DM) and −0.3% (SSC). More notable is the shape: uniformly negative and shrinking monotonically as m grows (DM from −1.67% to −0.67%, SSC from −1.59% to −0.12%), which agrees with the mechanistic expectation that the less noise is injected, the less of it is fitted away during retraining, and which was not obtained by any fitting; refitting is part but not all of that shape, since in a paired rerun a deviation of −0.97% (DM) and −0.92% (SSC) at $m = 1$ survives when the predictor is fixed. Supplementary S5 records that comparison and the paired construction behind the regularity.

A blinded stress test of the inversion. Four blinded targets were specified in advance, the m at which each lies having no measured result at the time of specification, inverted for the required replicate count and only then measured. What is inverted is Eq. (5) rather than Eq. (4), since the model here perceives the entire clean label and that baseline itself falls short of 1; Supplementary S5 gives the inverted expression, the four targets and their measured values. Two targets were met and two fell marginally short, all shortfalls within ±0.008. This is the same phenomenon as the negative deviation above: Eq. (4) accounts only for the structural attenuation caused by reference noise,

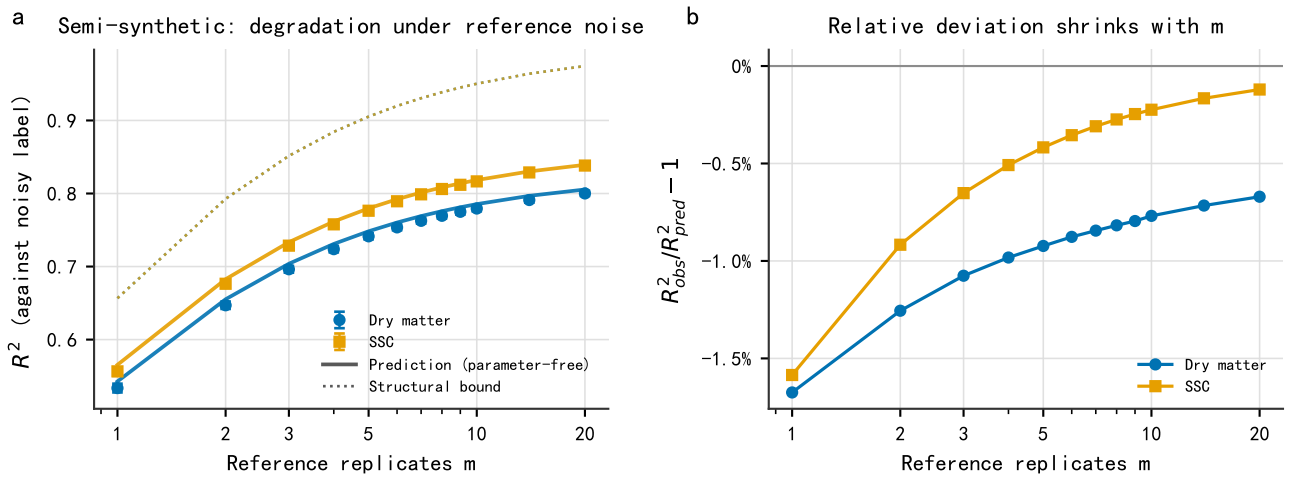


Figure 4: Semi-synthetic reference-noise degradation experiment. (a) Measured R^2 (points, error bars are seed-clustered bootstrap 95% CIs) against the prediction of Eq. (5) (solid) and the structural ceiling (dotted) as a function of the number of reference replicates m. (b) Shrinkage of the relative deviation $R^2_{\text{obs}}/R^2_{\text{pred}} - 1$ with m: uniformly negative and monotone, consistent with the mechanism that the extra retraining loss falls as the injected noise falls.

and the measured values fall slightly short of it, partly through the additional loss incurred when the model is retrained on noisy labels. The rule is therefore restated accurately as a lower bound on the required number of replicates, with an advisory margin, namely discount R^2_{clean} by 1% before inverting. That margin changes the required m only for the two targets that previously fell short: one rises from $m = 7$ to $m = 8$, where the measured value is 0.8062, turning a shortfall into a pass, and this is the one occasion on which the margin rule is directly verified by measurement; the other rises from $m = 9$ to $m = 11$, a level outside the grid, so no measured value exists, no interpolation is performed, and the neighbouring levels merely bracket it. The 1% figure is accordingly an empirical margin induced from these four points rather than a universal constant, and it must be recalibrated on other data.

The demonstration is not equivalent to validation on the apple data. What it establishes is narrower: when the spectral signal is strong enough, reference noise limits performance in the predicted way. It cannot conversely explain the weak apple performance, and scaling the apple variance ratio onto kiwifruit gives the demonstration a noise magnitude with an empirical basis rather than asserting that the kiwifruit reference determinations carry variability of that size. This section and the attenuation-ratio test of Section 3.2 bracket the same proposition from opposite sides: Section 3.2 uses real replicate reference determinations but a weak model ($R^2 \approx 0.06$–0.12), so its limitation is power, not validity, whereas here the model is strong but the replicates are synthetic and the injected noise is i.i.d. by construction, so the most fragile premise of Eq. (2) is supplied by us rather than tested. The weaknesses are orthogonal and the argument requires both sides. The failure mode most likely in practice is excluded by neither: the tissue sampled destructively may be the same tissue that the effective optical sampling depth probes, which would make reference noise correlated with what the model can perceive, and ruling that out needs a dataset carrying both real replicate reference determinations and a working spectral model (Section 4.3).

4. Discussion

4.1. Implications for reported calibration performance

The results do not imply that the high R^2 values reported in the literature are wrong. The split-leakage magnitude measured here, 1.41–1.46×, does not explain the gap to typical published values, and the apple data used here also perform poorly on the spectral side (Section 4.3), so this work cannot judge anyone else's model performance. The claim made is weaker and more robust: when the reference value is a single local determination, the ceiling of Eq. (2) holds for any predictor that can perceive only whole-fruit information, provided the mean-independence condition of Section 2.3 holds. That condition is what makes the statement model-free rather than linear-only, so the ceiling is a structural constraint independent of regressor, preprocessing and hyperparameters, and it must be stated precisely. If spectra and reference determinations are paired on the same face and local chemical differences do change the local spectrum, a model can in principle predict part of the biological content of e_{ij}, and Eq. (2) becomes a conditional ceiling for whole-fruit-level predictors only. The apple

data cannot distinguish the two cases, so the ceiling is asserted only under that condition.

The ceiling is not binding for every study. The apple data here measure $R^2 = 0.118$ on the whole-fruit mean target, so the ceiling to read it against is the five-face one, 0.820, not the single-face 0.477, and the measured value is far below either, so reference-value variability is not the binding constraint on that dataset's current performance. What does bind it is not established here: limited acquisition configuration is the hypothesis most consistent with the available diagnostics, but no experiment here varies that configuration and alternatives such as model class or parameter space were not individually excluded, so it is a candidate rather than a demonstrated cause. Whether a given study is bound depends on how far its measured R^2 sits from the ceiling implied by its own replicate determinations, while the attenuation diagnostic tests, on the same data, the premise behind that ceiling: that a whole-fruit model cannot perceive the per-face departures.

Two earlier results bear on how the ceiling should be read. It bounds the R^2 observable against the reference, not the accuracy a calibration attains: a calibration can perform better than the primary reference technique from which it was calibrated (DiFoggio, 1995), and the ceiling limits how much of that performance a single local determination can reveal. Reference-limited calibrations are also more than a theoretical possibility: on a public fruit dry-matter dataset, PLS prediction errors have been found close to the uncertainty of the reference technique (Olivieri, 2026). Recognising that situation on a study's own data requires replicate reference determinations, from which the ceiling can be computed and set against the measured R^2.

4.2. Practical implications for sampling and instrument deployment

On the reference side, the choice follows from the research target. If that target is whole-fruit quality, the reference value should be a multi-point mean and a lower bound on the number of replicates can be set with the design table of Section 3.5; if it is local quality, spectra and reference determinations must be paired at the same location and the result must not be evaluated against a whole-fruit label. Fixing the measurement position, as in the choice of a calibration position for pomelo (Xu et al., 2024), removes only a fixed positional gradient; in the apple data the fixed effect of face position explained just 0.95% of the within-fruit variability (Section 3.1), so most of that variance would remain in a reference taken at a single position. The attenuation diagnostic requires per-face replicate reference determinations on the same individual, so on one-fruit-one-label data it cannot be run and σ_f^2/σ_a^2 should be estimated first on a small subset. That pilot plays the part of the replication experiment in the theory of sampling, where replicate sampling estimates the total sampling-plus-analysis uncertainty and the sources contributing to it (Esbensen and Abu-Khalaf, 2022; Gajdoš Kljusurić et al., 2026). The practical yield is that a structural lower bound on the number of sampling faces per fruit can be set before an experiment starts, directing effort towards more reference replicates or a better model instead of towards tuning a model against a structurally unattainable target.

On the measurement side, an RMSEP obtained on a single instrument cannot show the error that arises only between instruments. In the kiwifruit data that error could be measured only pooled with scan/repositioning noise; the pooled component stayed nearly constant in absolute size while the total RMSEP fell, so its share of the error rose with model quality across the operating points compared (Section 3.3). Performance reports should therefore give the part of the error that changes with the instrument separately from the rest. Calibration-transfer methods target the correctable device main effect (Mishra et al., 2021a; Mishra and Passos, 2021; Yang et al., 2022; Li et al., 2022), but the fruit×device residual component cannot be separated from scan noise without technical replicates on the same device and is hard to remove by post-processing.

4.3. Limitations

The spectral side of the apple data performs poorly under the protocols evaluated here. Under fruit-grouped holdout the R^2 for whole-fruit mean SSC is only about 0.11, reproduced by two independently written implementations (0.118 and 0.109, overlapping intervals), which supports only insensitivity to implementation detail and not replication on independent data. Both are far below that dataset's five-face-mean ceiling of 0.820, so the conclusion is unaffected. Three diagnostics failed to support the candidate explanations: a row-wise random split gives only 0.169; clipping is rare in the archived spectra (Section 2.2), and excluding the bands at which more than 1% of spectra lie within 0.01 of either end of the scaled range, or restricting to 900–1001 nm, is worse than the full spectrum; and the same code and protocol give $R^2 = 0.827$ for DM and $R^2 = 0.861$ for SSC on the kiwifruit data. The single band most correlated with SSC is at 678 nm ($|r| = 0.154$), a colour–maturity proxy rather than direct sugar absorption. The reasonable attribution is twofold: reflectance geometry on intact fruit yields a surface-weighted sampling whose effective depth is typically shallow and varies with wavelength, geometry and fruit structure (Rodríguez-Ortega et al., 2023; Huang et al., 2024), whereas the refractometer measures juice from destructively sampled flesh, and 391–1001 nm covers only the very weak third overtone. Per-diagnostic figures are in Supplementary S5.

The source of the within-fruit variability cannot be fully decomposed. With no technical replicates on the same sampling face, genuine spatial chemical

heterogeneity cannot be separated from the variability of destructive sampling and preparation, and the term "within-fruit reference variability" is used throughout. Nothing in the practical conclusions is lost: provided face-level departures are approximately independent, which Section 3.2 tests, both sources decay as $1/m$ alike and Eqs. (2) and (4) are unchanged. Supplementary S5 gives that argument in full, with what the acquisition record does and does not contain.

The instrument component is a pooled term. With no technical replicates within a device, instrument× sample interaction and scan/repositioning noise are inseparable, so only the pooled term is reported and no claim is made about pure instrument repeatability. The decomposition is a sequential projection with neither REML nor rebalancing, and since the number of devices per fruit is not constant, unequal replication leaves the effect matrices no longer mutually orthogonal (de Figueiredo et al., 2023) and the two measurement shares order-dependent. The 1068–1137 nm columns are missing on all five devices, so the instrument budget covers only 402–1065 nm.

The two components come from different species; this is a division of labour, not mutual corroboration. They cannot validate each other, and the reason for placing them side by side is that no public dataset we could identify supplies both reference replicates and cross-instrument spectral replicates.

The three groups come from a single collection. All apple samples were purchased in a single 2025 acquisition from one wholesale market, and the record does not state whether the material constitutes a single batch; the origin labels are those declared by the merchants (Section 2.1 and Table 1). Since the ICC varies across the origin groups (0.335–0.615), the ceiling must be reported per collection. Reliability can vary with cultivar and season too, and modelling the variance components as heterogeneous is the established way to test that (Martinková et al., 2023); no such subgroup test was run, so the single ICC is an assumption rather than a finding.

Two preprocessing masks are computed before splitting. A paper about validation protocol should state where its own protocol falls short of the standard it argues for. The band mask behind the band-subset diagnostic and the missing-channel mask for the kiwifruit model are both built from the whole dataset and only then fed to the grouped holdouts. Both are target-blind, but under a strict no-clean-separation criterion (Kapoor and Narayanan, 2023) they remain decisions taken with held-out spectra in view, and a published route that removes the trade-off rather than bounding it (Galbo Engstrøm and Holm Jensen, 2025) was not taken here. For the kiwifruit mask the question can be settled rather than bounded, because the mask removes 0 of 222 wavelength columns and a fold-local mask is therefore necessarily identical for every split. For the band mask it was measured: recomputing the mask inside each outer training partition changes the retained wavelengths but leaves the band-subset conclusion unchanged: $R^2 = 0.081$ against 0.075 under the whole-data mask and 0.106 for the full spectrum in the same pass, so the whole-data mask biases the subset downwards, not upwards (Supplementary S5). Separately, one exploratory script selects a preprocessing method by held-out R^2; no number from it enters this paper, and it is flagged because the released code contains it. The affected quantities and the direct check are in Supplementary S5.

5. Conclusions

This study separated two variance components that are not the model's doing from the RMSEP of non-destructive fruit quality calibrations: reference-value variability, and between-instrument variability once a calibration is deployed across instruments. On the reference side, within-fruit variability made up 52.30% of apple SSC variance and, under the variance model of Section 2.3, puts the attainable R^2 ceiling at 0.477 (95% CI 0.436–0.515) for a single-face reference value and 0.820 (0.794–0.842) for the five-face mean. Across three origin groups bought in a single acquisition from one wholesale market the point estimates of the single-face ceiling vary between 0.335 and 0.615, a difference not formally tested, so the ceiling must be reported per collection rather than used as a species constant. Through the Spearman–Brown relation the reference-side result converts into a lower bound on the number of approximately independent reference replicates, $m \geq (\sigma_f^2/\sigma_a^2) \cdot R^2_{\max}/(1 - R^2_{\max})$, that can be set in advance once the variance ratio has been estimated on the batch at hand; for this collection a target ceiling of 0.8 needs at least five faces per fruit.

The variance-component model yields four groups totalling six parameter-free point predictions, whose measured values are all close to theory: the label-only baseline, for instance, has theoretical value 0.3462 against a measured 0.3456. Of these, exchangeability across faces was tested by permutation and the attenuation ratio against its bootstrap interval; the others carry no formal pass threshold. In a semi-synthetic experiment on the kiwifruit data, which injects reference noise scaled by the variance ratio measured on apples and so supplies that noise structure by construction rather than testing it on real fruit, the degradation of a retrained model agreed with the parameter-free prediction at all 12 finite replicate levels (median signed relative deviations of −0.9% and −0.3%). The measured values fell slightly short of that prediction, partly through the additional loss of retraining on noisy reference values, so the design rule, which accounts only for structural attenuation, is a lower bound; the 1% margin induced from four blinded targets applies to the

inverted degradation prediction and is empirical, not a universal constant.

On the measurement side, the pooled device×sample and scan/repositioning component corresponds to about 0.49 %DM of RMSEP in kiwifruit dry-matter prediction. That absolute quantity is approximately invariant between the two model operating points compared, the slope of its regression on R^2 having an interval that excludes zero while the fitted trend spans only 1.1% of the component's own magnitude, so under the strong model it accounts for 24.6% of the mean squared prediction error (24.7% under the ten-point scan). The design rule and the attenuation-ratio diagnostic both presuppose a leakage-free validation protocol: in the apple data the individual level was confirmed necessary, face-level random splitting inflating R^2 1.41–1.46-fold, whereas an apparent acquisition-day-level leakage in the time-series data was withdrawn after a control test that attributes the drop mainly to a target derived from the grouping variable, a genuine same-day shared-condition effect not being excluded.

For practice, a calibration for intact fruit should be read against a reference-limited ceiling rather than against unity, the number of faces sampled destructively per fruit can be planned from a pilot estimate of the variance ratio, and performance reports should separate the part of the error that changes with the instrument. The two components come from two datasets in a relation of division of labour, not mutual corroboration; how to calibrate both on the same batch deserves further study.

Data and code availability

The apple data are unpublished primary research material held by Xinjiang Agricultural University. The institution's data policy does not permit public release, so they are not deposited in a public repository; they may be requested from the corresponding author by email, and are shared at the institution's discretion under that policy. Every quantity derived from them is released in full, as set out below. The kiwifruit data are public and were used exactly as deposited: Kiwifruit NIR, figshare, v1, CC BY 4.0, doi:10.6084/m9.figshare.21747983.v1 (Wohlers, 2025), first described by Wohlers et al. (Wohlers et al., 2023). The third dataset is the public DeepHS-Fruit collection introduced by Varga et al. (Varga et al., 2021); its 2023 five-species release was used as distributed, from https://github.com/cogsys-tuebingen/deephs_fruit.

All analysis scripts, intermediate result workbooks, per-run records, figures and the pre-registered statistical protocol are released at https://github.com/2004lryan/020nir-reference-error-budget, together with a claim-by-claim map, so every statistic reported here except those of the cross-architecture replay is checkable without access to the raw data; the intermediate products of that replay were not retained, so it can only be repeated from the scripts and seeds, not rechecked quantity by quantity.

CRediT authorship contribution statement

Panlin Li: Methodology, Software, Formal analysis, Visualization, Writing – original draft. Yuchang Li: Investigation, Data curation, Validation. Yutong Feng: Investigation, Data curation. Longjie Li: Resources, Investigation. Ya Liu: Resources, Investigation. Hua Huang: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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